

Memory-Driven Dynamics of Self-Propelled Droplets

Experimental Analysis and PDE–SDE Modelling

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Outline

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- 2 Methodology
- 3 Memory-Driven Model
- 4 Numerical Simulation
- 5 Results and Discussion
 - Ensemble
- 6 Conclusion

Active Matter Systems

- Active matter consists of entities that consume energy and convert it into motion.
- Active systems are found in both living and synthetic systems.
- These systems exhibit persistence, collective behaviour, and non-equilibrium transport.

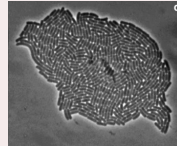
Motivating Question

What mechanism allows active droplets to remember where they have been?

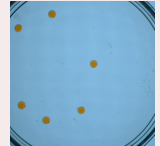
Examples of Active Matter



Bird flock



Bacteria

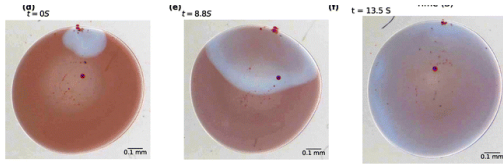


Active droplet

Image Sources

- Bird flock: *A murmuration of starlings at Gretna*, adapted from Walter Baxter (2011), Wikimedia Commons.
- Bacterial colony: Growing *E. coli* microcolony exhibiting active-nematic ordering, adapted from Dell'Arciprete *et al.*, *Nature Communications* (2018).
- Active droplet: Self-propelled Belousov–Zhabotinsky (BZ) droplet driven by chemical oscillations and Marangoni flow, Nonlinear Dynamics Laboratory, NITK.

Why Active Droplets?



BZ active droplet exhibiting internal chemical-wave dynamics. Adapted from Meshram and Shajahan (2025).

Key Features

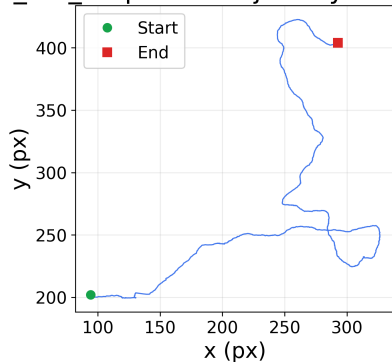
- Simple and controllable active matter system.
- Self-propelled through internally generated chemical activity.
- Continuously modifies its surrounding environment.
- Provides a natural platform for studying memory and feedback effects.

Hypothesis

The self-generated chemical field may retain information about previously visited locations and influence future motion.

Observation: Self-Avoiding Motion

for_2.1_droplets - Trajectory - Droplet 3



Observation

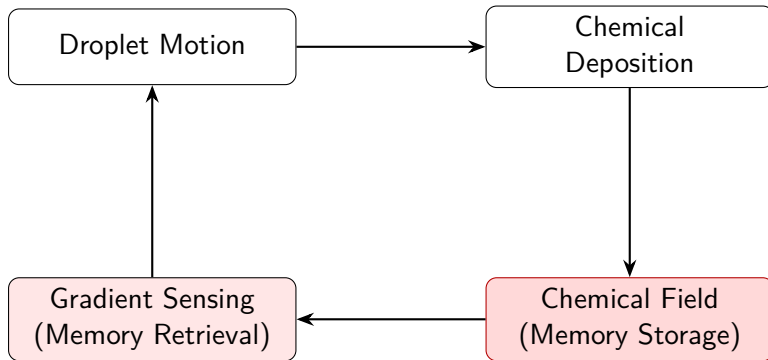
The droplet tends to avoid regions it has previously visited.

- Motion is not purely random.
- Trajectories exhibit self-avoidance.
- Past motion appears to influence future motion.

Research Question

What mechanism allows the droplet to remember where it has been?

A Possible Explanation: Memory



Hypothesis

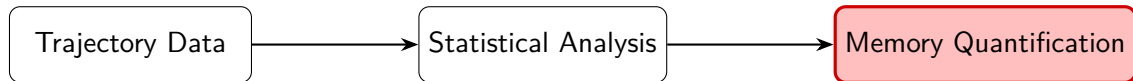
The droplet deposits a chemical trail that stores information about previously visited locations. By sensing gradients in this field, the droplet's future motion becomes influenced by its past motion.

Does the observed self-avoidance indicate memory-driven dynamics?

Questions

- 1 Does the droplet retain information about its past motion?
- 2 Can this memory be quantified from trajectory data?
- 3 Can a memory-driven model reproduce the observed behaviour?

How Do We Quantify Memory?



From observed trajectories, we extract statistical signatures of memory.

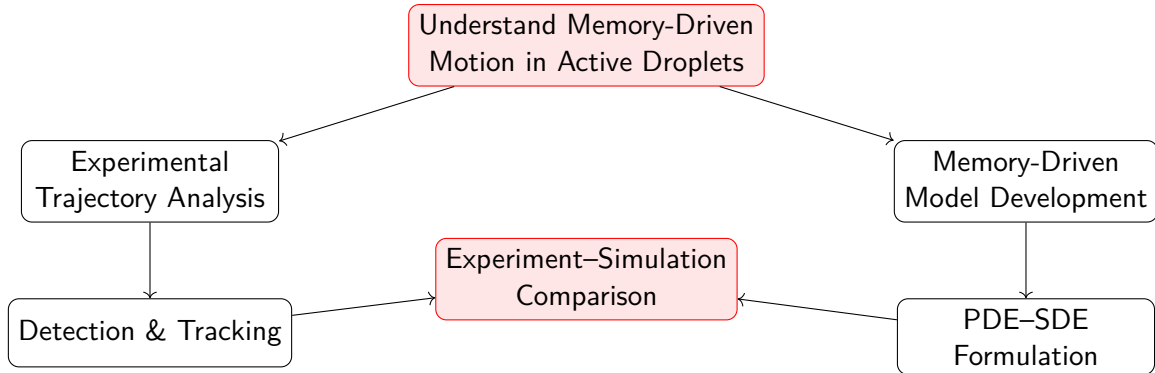
Transport Characterization

- MSD
How far does the droplet travel?
- Local MSD Exponent
Does transport change with time?

Memory Characterization

- Velocity Autocorrelation
How long is velocity remembered?
- Orientation Correlation
How long is direction remembered?

Research Strategy and Methodology

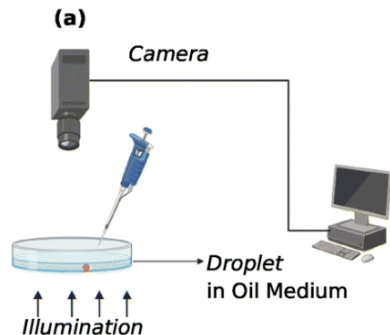


Objective

Combine experimental trajectory analysis with a memory-driven PDE-SDE model to identify and reproduce memory signatures in active droplet motion.

Experimental System

- Self-propelled Belousov–Zhabotinsky (BZ) droplets in a monoolein–squalane oil medium.
- Droplet diameter: ≈ 1.75 mm.
- Videos recorded using a CMOS camera at 5 fps.
- Spatial resolution: 0.06 mm per pixel.



Experimental setup for recording active droplet trajectories. Adapted from Meshram and Shajahan (2026).

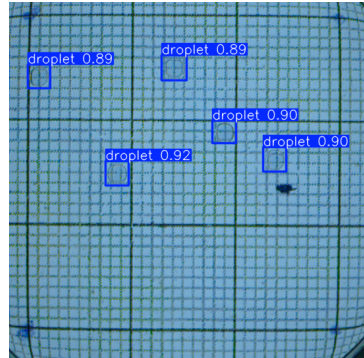
Vivek Bharat Meshram and T. K. Shajahan, *Soft Matter* (2026)

Trajectory Extraction

YOLOv8 Detection and Tracking

- Detect droplets in each frame.
- Assign persistent tracking IDs.
- Link detections across frames.
- Extract centroid coordinates.
- Reconstruct droplet trajectories.

Transfer learning using YOLOv8 (369 training images, 96 validation images, 50 epochs, resolution 640×640).

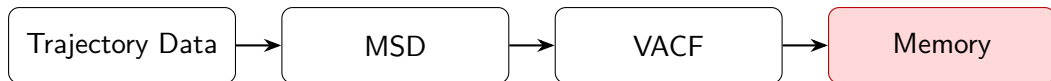


YOLOv8 detection and tracking output.
Bounding boxes and persistent tracking IDs
enable trajectory reconstruction across video frames.

Output

Tracked droplet trajectories used for transport and memory analysis.

From Trajectories to Memory Measures



Transport Measures

- Mean Squared Displacement (MSD)
- Local MSD Exponent

Memory Measures

- Velocity Autocorrelation Function (VACF)
- Orientation Correlation Function (OCF)

Transport Characterization

Mean Squared Displacement (MSD)

$$\text{MSD}(\tau) = \langle \|\mathbf{X}(t + \tau) - \mathbf{X}(t)\|^2 \rangle$$

- Measures spatial exploration.
- Identifies ballistic, diffusive, and anomalous transport.
- Used to estimate transport behavior.

Local MSD Exponent

$$\alpha(\tau) = \frac{d \log[\text{MSD}(\tau)]}{d \log(\tau)}$$

- Characterizes lag-dependent transport.
- $\alpha \approx 2$: Ballistic motion
- $\alpha \approx 1$: Normal diffusion
- $1 < \alpha < 2$: Superdiffusion

Physical Interpretation

MSD quantifies how far droplets travel, while the local exponent reveals how the transport regime evolves across timescales.

Memory Characterization

Velocity Autocorrelation Function (VACF)

$$C_v(\tau) = \frac{\langle \mathbf{v}(t) \cdot \mathbf{v}(t + \tau) \rangle}{\langle |\mathbf{v}(t)|^2 \rangle}$$

$$C_v(\tau) \sim e^{-\tau/\tau_p}$$

- Measures velocity memory.
- Provides persistence time τ_p .

Orientation Correlation Function (OCF)

$$C_\theta(\tau) = \langle \cos[\theta(t + \tau) - \theta(t)] \rangle$$

$$C_\theta(\tau) \sim e^{-\tau/\tau_r}$$

- Measures directional memory.
- Provides rotational persistence time τ_r .

Physical Meaning

These observables quantify how long information about past velocity and direction persists.

Memory and Non-Markovian Dynamics

Markovian Dynamics

$$P(X_{t+\Delta t} | X_t)$$

- Future depends only on the current state.
- No memory.
- Instantaneous response.

$$m\dot{v}(t) = -\gamma v(t) + \eta(t)$$

$$K(t) = \gamma\delta(t)$$

Non-Markovian Dynamics

$$P(X_{t+\Delta t} | X_t, X_{t-\tau}, \dots)$$

- Future depends on past history.
- Memory effects.
- Persistent correlations.

$$m\dot{v}(t) = -\int_0^t K(t-s)v(s) ds + \eta(t)$$

$$K(t) = K_0 e^{-t/\tau_m}$$

Motivation

Finite VACF and OCF decay indicate that information about past velocity and orientation persists for a finite time, motivating a non-Markovian description with memory kernels.

$$\frac{\partial c}{\partial t} = D \nabla^2 c + \alpha G_{R_t}(x - X_t)$$

- Droplet continuously deposits chemical material.
- Deposited material diffuses slowly.
- The field retains information about past positions.

Key Idea

The chemical field evolves more slowly than the droplet motion, allowing information about previously visited locations to persist.

Non-Markovian Droplet Model

Chemical Field (PDE)

$$\frac{\partial c}{\partial t} = D \nabla^2 c + \alpha G_{R_t}(\mathbf{x} - \mathbf{X}_t)$$

- Diffusion of chemical field.
- Moving droplet source.
- Stores memory.

Droplet Motion (SDE)

$$\eta_t \dot{\mathbf{X}}_t = -\mathbf{F}_{\text{chem}} + \sqrt{\sigma} \xi_t$$

$$\mathbf{F}_{\text{chem}} \propto \int_{\Omega} G_{R_t} \nabla c \, d\mathbf{x}$$

- Drift from chemical gradients.
- $-\nabla c$ gives self-repulsion.
- Noise drives fluctuations.

Feedback Loop

Motion $\rightarrow$ Chemical Field $\rightarrow$ Motion

Wenjun Chen et al., *Physical Review Letters* (2025)

Model Parameters and Notation

Model Equations

Chemical Field

$$\frac{\partial c}{\partial t} = D \nabla^2 c + \alpha G_{R_t}(\mathbf{x} - \mathbf{X}_t)$$

Droplet Motion

$$\eta_t \dot{\mathbf{X}}_t = -\frac{\beta R_t^3}{\delta^3} \int_{\Omega} G_{R_t}(\mathbf{x} - \mathbf{X}_t) \nabla c(\mathbf{x}, t) d\mathbf{x} + \sqrt{\sigma} \xi_t$$

Parameters

Symbol	Meaning
D	Diffusion coefficient
α	Source strength
R_t	Droplet radius
η_t	Friction coefficient
σ	Noise strength
δ	Interaction length scale
Ω	Computational domain

Key Idea

Past droplet positions are encoded in the chemical field and influence future motion through chemical-gradient sensing.

Wenjun Chen et al., *Physical Review Letters* (2025)

Finite Element Solution of the PDE

Weak Form

$$\int_{\Omega} \frac{\partial c}{\partial t} v \, dx + D \int_{\Omega} \nabla c \cdot \nabla v \, dx = \int_{\Omega} \alpha G_{R_t}(\mathbf{x} - \mathbf{X}_t) v \, dx$$

Boundary Condition

Reflective (no-flux) boundary:

$$\frac{\partial c}{\partial n} = 0 \quad \text{on } \partial\Omega$$

- Finite-element discretization.
- Implemented using FEniCS.

Numerical Integration of the SDE

Drift term

$$\mathbf{A}(\mathbf{X}_t, t) = -\frac{\beta R_t^3}{\gamma_t \delta^3} \int_{\Omega} G_{R_t}(\mathbf{x} - \mathbf{X}_t) \nabla c_t(\mathbf{x}) d\mathbf{x}$$

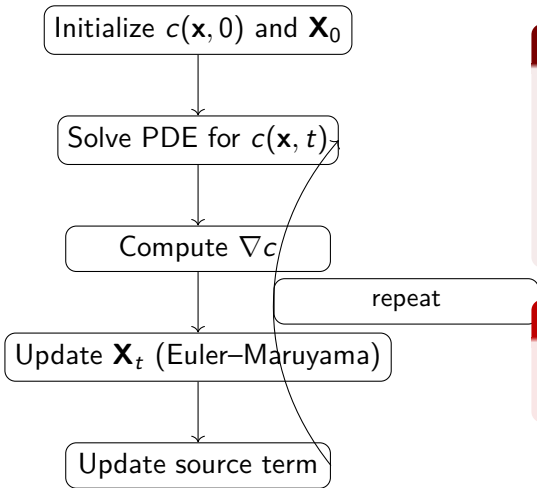
Euler–Maruyama Scheme

$$\mathbf{X}^{n+1} = \mathbf{X}^n + \mathbf{A}(\mathbf{X}^n, t_n) \Delta t + \frac{\sqrt{\sigma}}{\eta_t} \sqrt{\Delta t} \eta^n$$

$$\eta^n \sim \mathcal{N}(0, I)$$

- Deterministic drift from chemical gradients.
- Random fluctuations modeled by Gaussian noise.
- Generates stochastic droplet trajectories.

Simulation Workflow



Key Steps

- Evolve the chemical field using the diffusion PDE.
- Compute the gradient-induced drift.
- Update droplet position via Euler–Maruyama and refresh the source term.

Outcome

Simultaneous evolution of the chemical field and droplet trajectory generates the simulated paths.

Comparison Between Experiment and Simulation

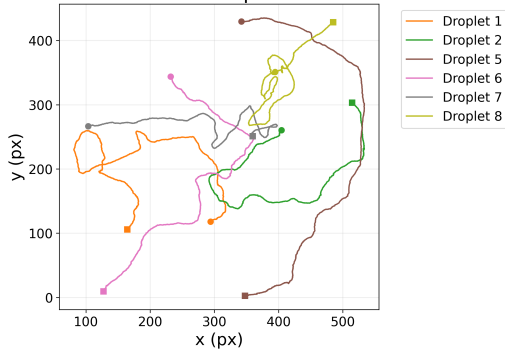
Ensemble Analysis Procedure

- MSD, VACF, and OCF were computed for each retained trajectory.
- Lag times were restricted to approximately 40% of the trajectory length to ensure reliable statistics.
- Ensemble observables were obtained by averaging over all retained droplets.
- MSD exponents were obtained from $\text{MSD}(\tau) \sim \tau^\alpha$ using the first 30% of the available lag-time range.
- Exponential fits to VACF and OCF were used to estimate characteristic persistence times τ_p and τ_r .
- Exponential fits $C_v(\tau) \sim e^{-\tau/\tau_p}$ and $C_\theta(\tau) \sim e^{-\tau/\tau_r}$ yielded τ_p and τ_r .

Ensemble Trajectory Comparison

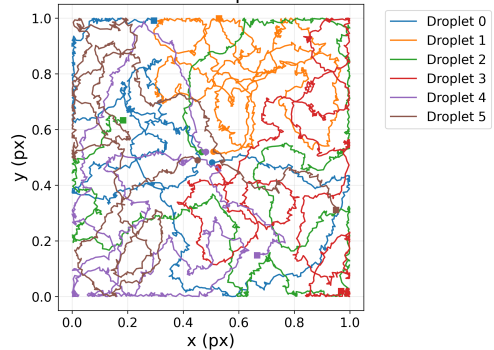
Experiment

for_2.3_droplets : Trajectory Comparison
N = 6 droplets



Simulation

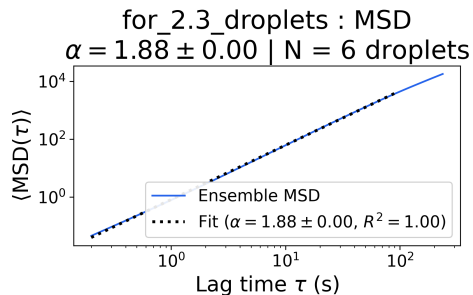
droplet_6_tracking : Trajectory Comparison
N = 6 droplets



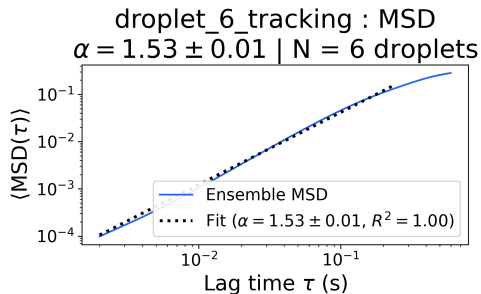
Persistent exploration and self-avoiding behaviour are observed in both datasets. The trajectory start and end positions are indicated by $\circ$ and $\blacksquare$, respectively.

Ensemble Mean-Squared Displacement

Experiment



Simulation



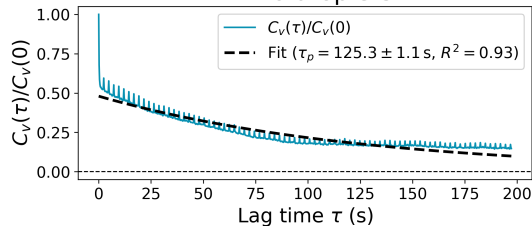
Key Observation

Both datasets exhibit superdiffusive transport ($\alpha > 1$), indicating persistent motion and departure from normal diffusion.

Ensemble Velocity Autocorrelation Function

Experiment

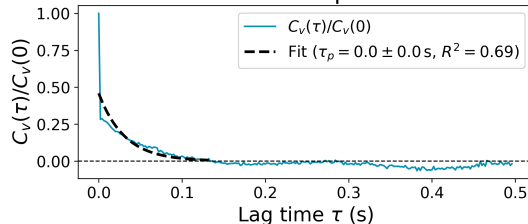
for_2.3_droplets : VACF
N = 6 droplets



$$\tau_p = 125.3 \pm 11.1 \text{ s}$$

Simulation

droplet_6_tracking : VACF
N = 6 droplets



$$\tau_p = 0.031 \pm 0.004$$

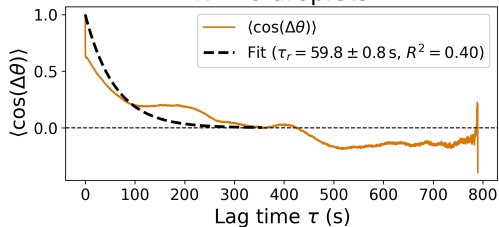
Key Observation

Both VACFs decay over a finite timescale, indicating persistent velocity correlations and finite translational memory.

Ensemble Orientation Correlation Function

Experiment

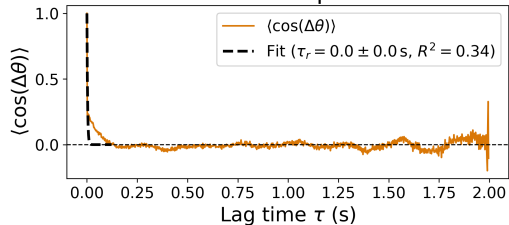
for_2.3_droplets : Orientation Correlation
N = 6 droplets



$$\tau_r = 59.8 \pm 0.8 \text{ s}$$

Simulation

droplet_6_tracking : Orientation Correlation
N = 6 droplets



$$\tau_r = 0.00305 \pm 0.00054$$

Key Observation

The orientation correlations decay gradually in both experiment and simulation, indicating finite directional persistence and rotational memory.

Summary of Ensemble Transport Parameters

Dataset	N	Frames / Iterations	α	τ_p (s)	τ_r (s)
2.1	4	3648	1.8345 ± 0.0019	61.83 ± 0.36	13.49 ± 0.34
2.2	6	3363	1.6482 ± 0.0053	17.14 ± 0.56	3.96 ± 0.26
2.3	6	3951	1.8776 ± 0.0012	125.25 ± 1.09	59.78 ± 0.78
2.4	3	5267	1.7025 ± 0.0049	28.40 ± 0.22	32.18 ± 0.35
2&4 μL	4	5526	1.5592 ± 0.0066	14.32 ± 0.16	10.83 ± 0.20
Simulation	6	1000	1.5272 ± 0.0088	0.0310 ± 0.0036	0.00305 ± 0.00054
Simulation (Scaled)	6	1000	1.5272 ± 0.0088	14.32 ± 1.66	1.41 ± 0.25

Simulation persistence times were converted using $t_{\text{scale}} = \tau_p^{\text{exp}} / \tau_p^{\text{sim}} \approx 462$.

Key Observation

The model reproduces the superdiffusive transport exponent ($\alpha > 1$) and exhibits finite velocity and directional persistence, consistent with memory-driven dynamics.

Experiment–Simulation Comparison

Feature	Experiment	Simulation
Self-avoidance	✓	✓
Superdiffusion ($\alpha > 1$)	✓	✓
VACF decay	✓	Partial
OCF decay	✓	Partial
Translational memory	✓	Partial
Rotational memory	✓	Partial
Non-Markovian behaviour	✓	Qualitative

Limitation: The simulation reproduces the qualitative decay of the VACF and OCF, but the persistence times (τ_p, τ_r) differ substantially from the experimental values.

Conclusion

The PDE–SDE model captures the key transport and memory signatures (self-avoidance, superdiffusion, and finite correlation decay), but does not quantitatively reproduce the experimental persistence times.

Conclusions

Main Findings

- Experimental droplet trajectories exhibit persistent self-avoiding motion.
- MSD analysis indicates superdiffusive transport ($\alpha > 1$).
- VACF and OCF reveal finite translational (τ_p) and rotational (τ_r) memory.
- A memory-driven PDE–SDE model was developed based on droplet–chemical field interactions.
- The model reproduces the key qualitative transport and memory signatures observed experimentally.

Key Limitation

The model reproduces the qualitative decay of the VACF and OCF, but significantly underestimates the persistence times (τ_p, τ_r) measured experimentally.

Outlook and Future Work

Physical Interpretation

The results support the hypothesis that active droplet motion is influenced by memory effects arising from self-generated chemical fields, leading to history-dependent dynamics.

Future Directions

- Calibrate model parameters for quantitative agreement.
- Investigate memory-kernel based descriptions (e.g., Generalized Langevin Equation).
- Extend the framework to interacting droplets and collective behaviour.
- Explore long-time transport and anomalous diffusion regimes.

Long-Term Goal

Develop predictive memory-driven models for active matter systems that couple stochastic motion with evolving environmental fields.

- Chen, Wenjun et al. (2025). “Evolving Motility of Active Droplets Is Captured by a Self-Repelling Random Walk Model”. In: *Physical Review Letters* 134.1, p. 018301. DOI: 10.1103/PhysRevLett.134.018301.
- Meshram, Vivek Bharat and T. K. Shajahan (2026). “Excitability induced binary interactions in self-propelled Belousov–Zhabotinsky droplets”. In: *Soft Matter* 22 (15), pp. 2938–2947. DOI: 10.1039/D5SM01187F. URL: <http://dx.doi.org/10.1039/D5SM01187F>.